

Method for Voltage Sag Disturbance Source Location by the Real Current Component

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Abstract—Voltage sag has become the most concerned about power quality problems. Voltage sag source detection for power quality fault detection, diagnosis and to ease the formulation of strategy is necessary. This paper proposes a new method about voltage sag disturbance source location based on real current component. The product of the RMS current and the power factor angle at the monitoring point is employed before and after the continuous period of time for fault. A graph of this product against time is plotted. The real current component method locates the disturbance source by judging the polar of the real current. This method improves the detection accuracy, speed up the detection rate and improves the real-time algorithm to achieve fast positioning accuracy of the disturbance. The simulation results verify the validity of the method.

key words—Voltage sag; Disturbance location; The real current component

I. INTRODUCTION

Voltage sag, also known as voltage drop, fall dip, or voltage dips and so on. IEEE standard is a temporary decrease in the RMS voltage magnitude between 0.1 and 0.9 p.u. and with duration of mostly less than 1 s. With the development of power electronics and automatic control technology, programmable logic devices, speed drives and other devices become increasingly sensitive to voltage sag, and these devices are used more widely, so for industrial users, the voltage sag has become a major power quality problem. Meanwhile, for the electricity sector, the customer complaints and an increase in economic disputes caused by the voltage sag, will affect its competitiveness in the environment of electricity market [1]. Voltage sag source detection for power quality fault detection, diagnosis and to ease the formulation of strategy is necessary [2].

In present, according to the study about the voltage sag source location, some of the methods can be generally summarized into the following three: (1) a method based on the disturbance power and disturbance energy [3]. This method using the disturbance power and disturbance energy detection to determine disturbance source is based on the concept that the disturbance power and disturbance energy changes caused by disturbance. But it does not have mathematical analysis algorithm, and the reliability of this method will reduce when the disturbance power and disturbance energy result does not match [4]; (2) a method based on line protection, such as using the distance and impedance relay. Most of the transmission line will be

installed the distance relay for the need of protection. This approach uses the change of impedance and phase angle before and after the voltage sag to determine the voltage sag source [5]. But in the radiation network, this approach has some limitations; (3) the methods using artificial intelligence computing, wavelet transform and others [6]. These methods have not really used to the traditional voltage sag positioning.

This paper proposes a method which is based on the directional relay concept using the power factor angle to locate the disturbing source. In the method, magnitude of currents and phase angles of voltages and currents are measured at the measuring point. The current magnitude is then multiplied with the cosine of the power factor angle and the product is then plotted against time. The product polarity is used to indicate the direction of voltage sag source either it is from upstream or downstream relative to the measuring point. A positive polarity of the product at the beginning of the sag duration will indicate the voltage sag source is from downstream. On the other hand, a negative polarity of the product at the beginning of the sag duration will indicate that the voltage sag source is from upstream. Simulation results show the adaptability of this method.

II. THE REAL CURRENT COMPONENT CONCEPT

An equivalent circuit representing the two-source system is considered as shown in Fig. 1. Consider a fault at X and two monitoring points M1 and M2.

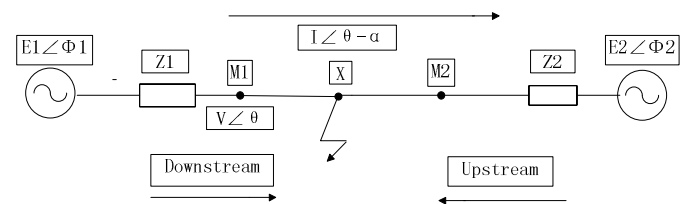


Figure 1. A two-source system for the voltage sag source location analysis

For a fault at X, using Kirchoff's second law, the voltage at M1 will be

$$\tilde{V} = \tilde{E}_1 - \tilde{I}\tilde{Z}_1 \quad (1)$$

Where $\tilde{V}$ and $\tilde{I}$ are the voltage and current measured directly at M1, respectively. Multiply on both sides of Eq. (1) by I^* will lead to

$$\tilde{V}I^* = \tilde{E}_1 \tilde{I}^* - I^2 \tilde{Z}_1 \quad (2)$$

Extracting the real part of Eq. (2) will lead to

$$VI \cos(\theta - \alpha) = E_1 I \cos(\phi_1 - \alpha) - I^2 R \quad (3)$$

Where θ and α are the phase angles for voltage and current at the measuring point M1, respectively. The phase difference $(\theta - \alpha)$ is the power factor angle at the monitoring point.

From (3), for the case of monitoring at a point before the fault X, i.e. M1, the active power flows from E1 to X, $I \cos(\theta - \alpha) > 0$. The fault is said to occur at downstream location as seen from the point M1 and the voltage sag source is seen coming from downstream. This current is in the same direction as the pre-fault current from E1 to E2. In the case of monitoring at point after the fault X, i.e. M2, the current $I \cos(\theta - \alpha) < 0$ because the current already flows to ground at the fault point and this current is seen as current flowing from E2 to X. The fault is said to occur at upstream location as seen from the monitoring point M2 and the voltage sag source is seen coming from upstream.

In the proposed method the value of fault impedance is taken into consideration because if a fault impedance, Z_f is less than Z_2 , then no current is seen flowing from E_2 since all current flows through the fault path, from the source E_1 to Z_f . However, at the beginning of voltage sag, the current can be significantly higher than the steady-state current due to a sudden change in electrical condition. Therefore a more suitable indicator chosen for voltage sag source location will be based on the direction of current at the beginning of a fault. The condition for the proposed method is that if at the beginning of a fault, $I \cos(\theta - \alpha) > 0$, the source of voltage sag is from downstream. On the other hand, if $I \cos(\theta - \alpha) < 0$, the source of voltage sag is from upstream.

The concept of the direction of real current components

used for voltage sag source location can both be applied to a single and more source system.

III. LOCATION STEPS OF THE REAL CURRENT COMPONENT METHOD

step1: Built a condition model. Create voltage sag conditions by simulating different short-circuit faults at the fixed points.

step2: Samples the magnitude and phase of voltage and current at pre-fault and during fault times, then calculate the current RMS.

step3: Calculate the values of $I \cos(\theta - \alpha)$, then plot coordinates of $I \cos(\theta - \alpha)$ against time of a few cycles of pre-fault and during fault durations.

step4: Check the polarity of $I \cos(\theta - \alpha)$ at the beginning of fault. If it is positive, the source of voltage sag is from downstream. On the other hand, if it is negative, the source of voltage sag is from upstream.

IV. SIMULATIONS

In this section, one-source and two source system are established, and result of balanced and unbalanced fault for voltage sag is simulated. Real current component method (RCM) used to locate is compared with the result located by slope method (SM), to verify its adaptability and superiority

A. Location of voltage sag source caused by one-source system

Figure 2 shows a one-source radial power distribution network model which consists of three fault points F1, F2, F3 and three monitoring points M1, M2, M3. MATLAB/Simulink is used to simulate the line faults to create voltage sag conditions.

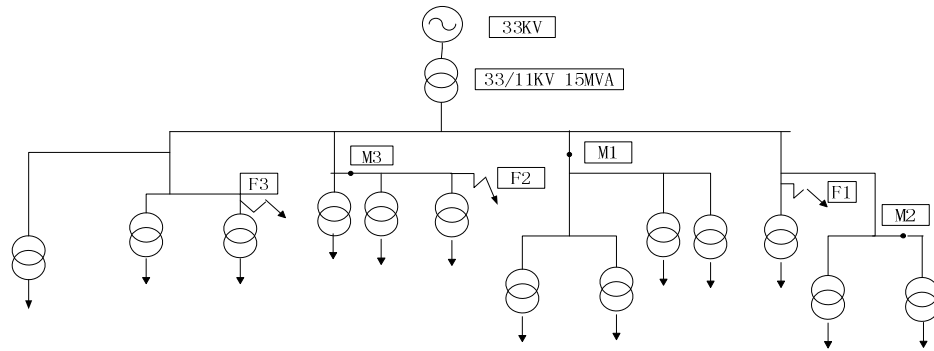


Figure 2. Test system for one-source voltage sag source location

1) Balanced fault on line

First the voltage sag caused by three-phase short circuit faults at F1, F2 and F3 was analyzed. Figure (a)-(c) shows the graph of location waveform measured made at M1, M2 and M3 by the real current component method, respectively. Figure (e)-(g) shows the location waveform made at M1, M2 and M3 by the slope method [4].

In figure 3 (a), it can be seen that real current rises sharply from the beginning of sag, shows the voltage sag source is located downstream of M1 point. Figure (b) and (c) shows that the real current sharp declines at the beginning of voltage sag, so we can determine the voltage sag source caused by three-phase short circuit fault of the F1 points is located upstream of M2, M3.

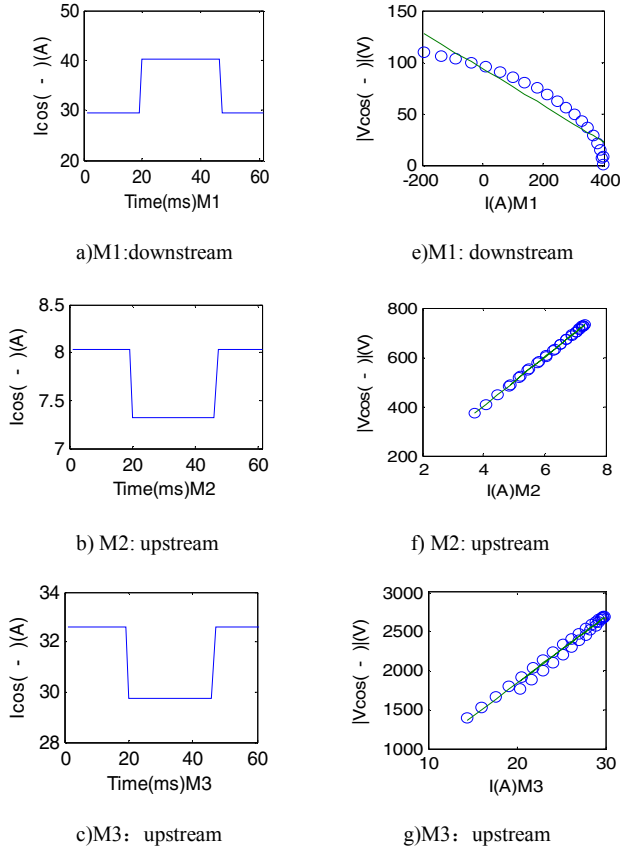


Figure 3. Fault at F1 for monitoring at M1, M2 and M3 using the RCM and SM

From Figure 3 (e), we can see that the slope of the line fitted by the point $(I, |V \cos \theta_2|)$ is negative, which indicating the voltage sag source is located downstream of M1 point [4]; On the other hand, the line's slope in figure (f) and (g) is positive, indicating the voltage sag source caused by three-phase short circuit fault at the F1 points is located

upstream of M2, M3.

Location of voltage sag source caused by balance fault at M1, M2, and M3 was analyzed respectively, and the two location methods reached unified conclusions as follows:

TABLE I. BALANCED FAULT BUS DETAILS FOR ONE-SOURCE SYSTEM

Fault points	Monitoring points	
	Downstream fault	Upstream fault
F1	M1	M2,M3
F2	M3	M1,M2
F3	—	M1,M2,M3

2) Unbalanced fault on line

In this section voltage sag caused by asymmetric fault (single phase fault / SLG, two-phase fault / DLG) at F1, F2 and F3 was simulated

Taking F1 for example, we can see from figure 2 that the real current component method can get the exact location of F1, no matter what kind of fault occurred that caused voltage sag. We can also get exact results at F2 and F3 by the real current component method and the same results with the slope method.

TABLE II. UNBALANCED FAULT BUS DETAILS FOR ONE-SOURCE SYSTEM

Fault points	Monitoring points	
	Downstream fault	Upstream fault
SLG	M1	M2,M3
DLG	M1	M2,M3

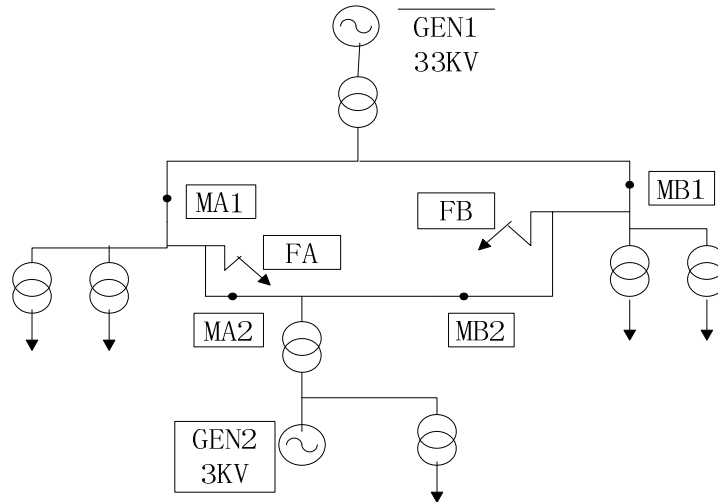


Figure 4. Test system for two-source voltage sag source location

B. Location of voltage sag caused by two source system

Two-source system is established as figure 4, which consisted of 2 fault point F1 F2, and 4 monitoring point MA1, MA2, MB1 and MB2. Two kinds of faults are considered, balance and unbalance faults that will cause voltage sag.

1) Balanced fault on line

First the voltage sag caused by three-phase short circuit faults at FA and FB was simulated. Figure (a)-(c) shows the graph of location waveform measured made at MB1, MB2 and MA1 by the real current component method respectively. Figure (e)-(g) show the location waveform made at MB1, MB2 and MA1 by the slope method.

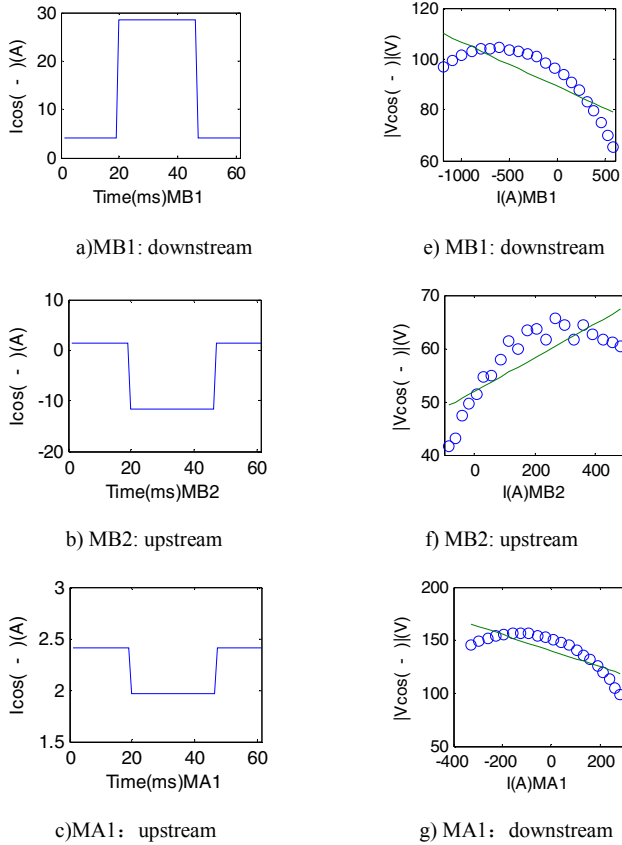


Figure 5. Fault at FB for monitoring at M1, M2 and M3 using the RCM and SM

Figure 5 shows that results from the real current component method are identical to the actual event. The result from the slope method at MA1 is upstream which contradict with the actual event in which the voltage sag source is actually from upstream.

By analyzing the location of voltage sag caused by two balance points from MA1, MA2, MB1 and MB2 with the real current component method, we can get conclusions as follows:

Fault points	Monitoring points	
	Downstream fault	Upstream fault
FA	MA1	MA2,MB1,MB2
FB	MB1	MA1,MA2,MB2

2) Unbalanced fault on line

In this section voltage sag caused by unbalance fault (single phase fault, two-phase fault) at FA and FB was simulated. By location analysis, we can see that the real current component method can give exact location for dual power system, while the slope method can not turn out exact location at some points

V. CONCLUSIONS

This paper proposed a new method detected the voltage sag at the monitoring points using the real current to locate the disturbing source. By location analysis, we found that the real current component method is valid for both balance or unbalance fault occurred in one-source or two source system or even power lines. This method only needs to calculate the value of the beginning time $I \cos(\theta - \alpha)$ for voltage sag, and then fit these points to get a picture of waveform relative to time, no need to fit linearly, and its less calculation can shortens the location time.

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TABLE III. BALANCED FAULT BUS DETAILS FOR TWO-SOURCE SYSTEM